GENERAL INFORMATION

DESCRIPTION AND OPERATION

STEERING ANGLE SENSOR MODULE (SASM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Steering Angle Sensor Module (SASM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.
For additional information, refer to: Braking Control System (206-11 Brake

Controls, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1B00-1C	Steering Angle Sensor - Circuit voltage out of range	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-49	Steering Angle Sensor - Internal electronic failure	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-61	Steering Angle Sensor - Signal calculation failure	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-62	Steering Angle Sensor - Signal compare failure	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-64	Steering Angle Sensor - Signal plausibility failure	▪ Signal plausibility failure ▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-81	Steering Angle Sensor - Invalid serial data received	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-85	Steering Angle Sensor - Signal above allowable range	▪ Steering angle sensor module/clockspring incorrectly aligned ▪ Steering angle sensor module	▪ Check if the steering angle sensor module or the clockspring has been removed and refitted incorrectly ▪ If fault persists, check and install a new steering angle sensor

		▪ internal failure	module as required
C1B00-92	Steering Angle Sensor - Performance or incorrect operation	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-96	Steering Angle Sensor - Component internal failure	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
C1B00-97	Steering Angle Sensor - Component or system operation obstructed or blocked	▪ Steering angle sensor module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new steering angle sensor module
U0001-88	High Speed CAN Communication Bus - Bus off	▪ Chassis high speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the chassis high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Steering angle sensor module power or ground circuit open circuit, high resistance ▪ Chassis high speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Incorrect steering angle sensor module installed ▪ Steering angle sensor module	▪ Refer to the electrical circuit diagrams and check the steering angle sensor module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the chassis high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check correct steering angle sensor module is installed ▪ Using the manufacturer approved diagnostic system, clear the DTCs

		internal failure	and retest. If the fault persists, install a new steering angle sensor module
U3003-62	Battery Voltage - Signal compare failure	▪ Mismatch between the voltage at the steering angle sensor module and the voltage value broadcast on the CAN bus	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the steering angle sensor module power and ground circuits for open circuit, high resistance